

$$16. \iint_R y \sin(x+y) dA, \quad R: -\pi \leq x \leq 0, \quad 0 \leq y \leq \pi$$

$$17. \iint_R e^{x-y} dA, \quad R: 0 \leq x \leq \ln 2, \quad 0 \leq y \leq \ln 2$$

$$18. \iint_R xye^{xy^2} dA, \quad R: 0 \leq x \leq 2, \quad 0 \leq y \leq 1$$

$$19. \iint_R \frac{xy^3}{x^2+1} dA, \quad R: 0 \leq x \leq 1, \quad 0 \leq y \leq 2$$

$$20. \iint_R \frac{y}{x^2y^2+1} dA, \quad R: 0 \leq x \leq 1, \quad 0 \leq y \leq 1$$

In Exercises 21 and 22, integrate f over the given region.

21. **Square** $f(x, y) = 1/(xy)$ over the square $1 \leq x \leq 2, 1 \leq y \leq 2$

22. **Rectangle** $f(x, y) = y \cos xy$ over the rectangle $0 \leq x \leq \pi, 0 \leq y \leq 1$

$$z = f(x, y)$$

Volume Beneath a Surface

23. Find the volume of the region bounded above by the paraboloid $z = x^2 + y^2$ and below by the square $R: -1 \leq x \leq 1, -1 \leq y \leq 1$.

24. Find the volume of the region bounded above by the elliptical paraboloid $z = 16 - x^2 - y^2$ and below by the square $R: 0 \leq x \leq 2, 0 \leq y \leq 2$.

25. Find the volume of the region bounded above by the plane $z = 2 - x - y$ and below by the square $R: 0 \leq x \leq 1, 0 \leq y \leq 1$.

26. Find the volume of the region bounded above by the plane $z = y/2$ and below by the rectangle $R: 0 \leq x \leq 4, 0 \leq y \leq 2$.

27. Find the volume of the region bounded above by the surface $z = 2 \sin x \cos y$ and below by the rectangle $R: 0 \leq x \leq \pi/2, 0 \leq y \leq \pi/4$.

28. Find the volume of the region bounded above by the surface $z = 4 - y^2$ and below by the rectangle $R: 0 \leq x \leq 1, 0 \leq y \leq 2$.

15.2 Double Integrals over General Regions

In this section we define and evaluate double integrals over bounded regions in the plane which are more general than rectangles. These double integrals are also evaluated as iterated integrals, with the main practical problem being that of determining the limits of integration. Since the region of integration may have boundaries other than line segments parallel to the coordinate axes, the limits of integration often involve variables, not just constants.

Double Integrals over Bounded, Nonrectangular Regions

To define the double integral of a function $f(x, y)$ over a bounded, nonrectangular region R , such as the one in Figure 15.8, we again begin by covering R with a grid of small rectangular cells whose union contains all points of R . This time, however, we cannot exactly fill R with a finite number of rectangles lying inside R , since its boundary is curved, and some of the small rectangles in the grid lie partly outside R . A partition of R is formed by taking the rectangles that lie completely inside it, not using any that are either partly or completely outside. For commonly arising regions, more and more of R is included as the norm of a partition (the largest width or height of any rectangle used) approaches zero.

Once we have a partition of R , we number the rectangles in some order from 1 to n and let ΔA_k be the area of the k th rectangle. We then choose a point (x_k, y_k) in the k th rectangle and form the Riemann sum

$$S_n = \sum_{k=1}^n f(x_k, y_k) \Delta A_k.$$

As the norm of the partition forming S_n goes to zero, $\|P\| \rightarrow 0$, the width and height of each enclosed rectangle goes to zero and their number goes to infinity. If $f(x, y)$ is a continuous function, then these Riemann sums converge to a limiting value, not dependent on any of the choices we made. This limit is called the **double integral** of $f(x, y)$ over R :

$$\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(x_k, y_k) \Delta A_k = \iint_R f(x, y) dA.$$

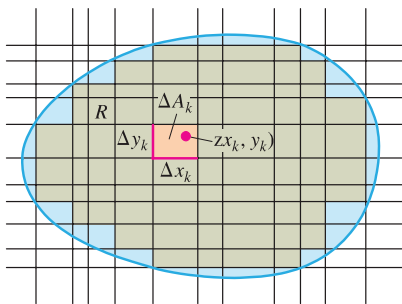


FIGURE 15.8 A rectangular grid partitioning a bounded nonrectangular region into rectangular cells.

The nature of the boundary of R introduces issues not found in integrals over an interval. When R has a curved boundary, the n rectangles of a partition lie inside R but do not cover all of R . In order for a partition to approximate R well, the parts of R covered by small rectangles lying partly outside R must become negligible as the norm of the partition approaches zero. This property of being nearly filled in by a partition of small norm is satisfied by all the regions that we will encounter. There is no problem with boundaries made from polygons, circles, ellipses, and from continuous graphs over an interval, joined end to end. A curve with a “fractal” type of shape would be problematic, but such curves arise rarely in most applications. A careful discussion of which type of regions R can be used for computing double integrals is left to a more advanced text.

Volumes

If $f(x, y)$ is positive and continuous over R , we define the volume of the solid region between R and the surface $z = f(x, y)$ to be $\iint_R f(x, y) dA$, as before (Figure 15.9).

If R is a region like the one shown in the xy -plane in Figure 15.10, bounded “above” and “below” by the curves $y = g_2(x)$ and $y = g_1(x)$ and on the sides by the lines $x = a, x = b$, we may again calculate the volume by the method of slicing. We first calculate the cross-sectional area

$$A(x) = \int_{y=g_1(x)}^{y=g_2(x)} f(x, y) dy$$

and then integrate $A(x)$ from $x = a$ to $x = b$ to get the volume as an iterated integral:

$$V = \int_a^b A(x) dx = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx. \quad (1)$$

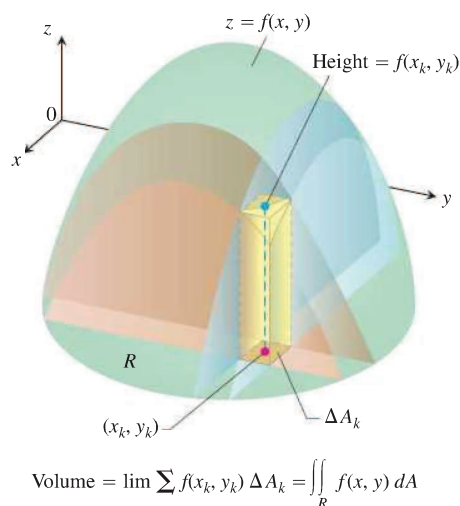


FIGURE 15.9 We define the volumes of solids with curved bases as a limit of approximating rectangular boxes.

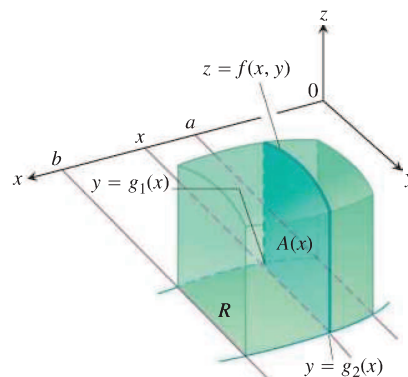


FIGURE 15.10 The area of the vertical slice shown here is $A(x)$. To calculate the volume of the solid, we integrate this area from $x = a$ to $x = b$:

$$\int_a^b A(x) dx = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx.$$

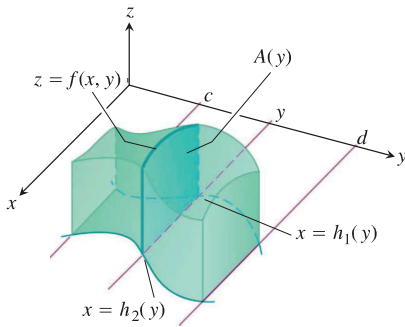


FIGURE 15.11 The volume of the solid shown here is

$$\int_c^d A(y) dy = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy.$$

For a given solid, Theorem 2 says we can calculate the volume as in Figure 15.10, or in the way shown here. Both calculations have the same result.

Similarly, if R is a region like the one shown in Figure 15.11, bounded by the curves $x = h_2(y)$ and $x = h_1(y)$ and the lines $y = c$ and $y = d$, then the volume calculated by slicing is given by the iterated integral

$$\text{Volume} = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy. \quad (2)$$

That the iterated integrals in Equations (1) and (2) both give the volume that we defined to be the double integral of f over R is a consequence of the following stronger form of Fubini's Theorem.

THEOREM 2—Fubini's Theorem (Stronger Form) Let $f(x, y)$ be continuous on a region R .

1. If R is defined by $a \leq x \leq b$, $g_1(x) \leq y \leq g_2(x)$, with g_1 and g_2 continuous on $[a, b]$, then

$$\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx.$$

2. If R is defined by $c \leq y \leq d$, $h_1(y) \leq x \leq h_2(y)$, with h_1 and h_2 continuous on $[c, d]$, then

$$\iint_R f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy.$$

EXAMPLE 1 Find the volume of the prism whose base is the triangle in the xy -plane bounded by the x -axis and the lines $y = x$ and $x = 1$ and whose top lies in the plane

$$z = f(x, y) = 3 - x - y.$$

Solution See Figure 15.12. For any x between 0 and 1, y may vary from $y = 0$ to $y = x$ (Figure 15.12b). Hence,

$$\begin{aligned} V &= \int_0^1 \int_0^x (3 - x - y) dy dx = \int_0^1 \left[3y - xy - \frac{y^2}{2} \right]_{y=0}^{y=x} dx \\ &= \int_0^1 \left(3x - \frac{3x^2}{2} \right) dx = \left[\frac{3x^2}{2} - \frac{x^3}{2} \right]_{x=0}^{x=1} = 1. \end{aligned}$$

When the order of integration is reversed (Figure 15.12c), the integral for the volume is

$$\begin{aligned} V &= \int_0^1 \int_y^1 (3 - x - y) dx dy = \int_0^1 \left[3x - \frac{x^2}{2} - xy \right]_{x=y}^{x=1} dy \\ &= \int_0^1 \left(3 - \frac{1}{2} - y - 3y + \frac{y^2}{2} + y^2 \right) dy \\ &= \int_0^1 \left(\frac{5}{2} - 4y + \frac{3}{2}y^2 \right) dy = \left[\frac{5}{2}y - 2y^2 + \frac{y^3}{2} \right]_{y=0}^{y=1} = 1. \end{aligned}$$

The two integrals are equal, as they should be. ■

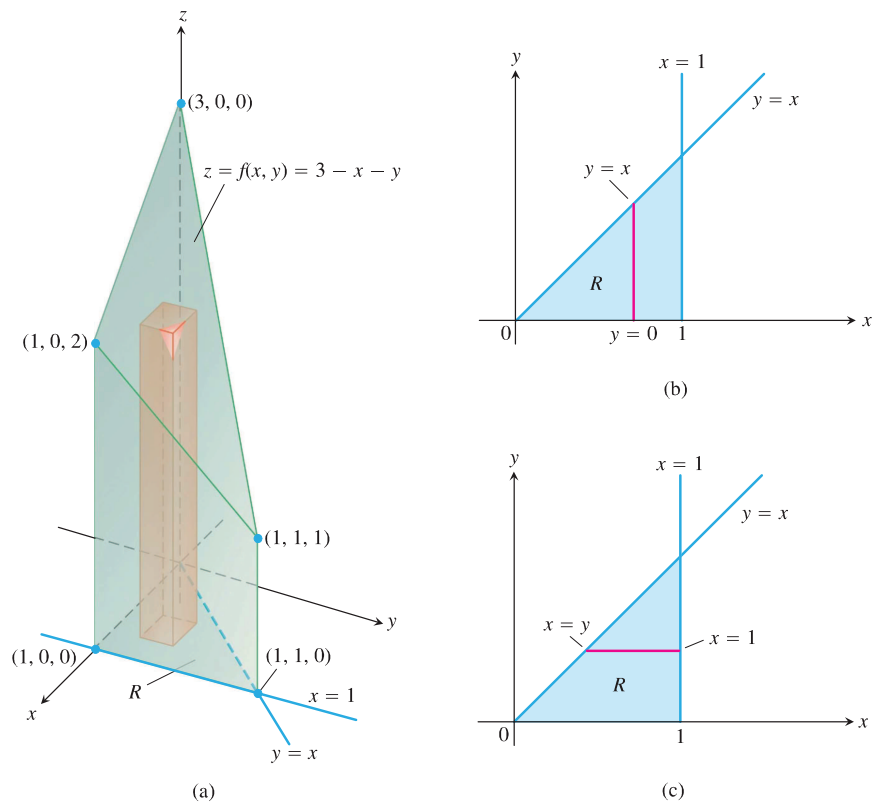


FIGURE 15.12 (a) Prism with a triangular base in the xy -plane. The volume of this prism is defined as a double integral over R . To evaluate it as an iterated integral, we may integrate first with respect to y and then with respect to x , or the other way around (Example 1). (b) Integration limits of

$$\int_{x=0}^{x=1} \int_{y=0}^{y=x} f(x, y) \, dy \, dx.$$

If we integrate first with respect to y , we integrate along a vertical line through R and then integrate from left to right to include all the vertical lines in R . (c) Integration limits of

$$\int_{y=0}^{y=1} \int_{x=y}^{x=1} f(x, y) \, dx \, dy.$$

If we integrate first with respect to x , we integrate along a horizontal line through R and then integrate from bottom to top to include all the horizontal lines in R .

Although Fubini's Theorem assures us that a double integral may be calculated as an iterated integral in either order of integration, the value of one integral may be easier to find than the value of the other. The next example shows how this can happen.

EXAMPLE 2 Calculate

$$\iint_R \frac{\sin x}{x} \, dA,$$

where R is the triangle in the xy -plane bounded by the x -axis, the line $y = x$, and the line $x = 1$.

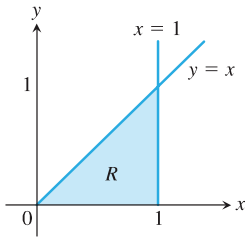


FIGURE 15.13 The region of integration in Example 2.

Solution The region of integration is shown in Figure 15.13. If we integrate first with respect to y and then with respect to x , we find

$$\begin{aligned} \int_0^1 \left(\int_0^x \frac{\sin x}{x} dy \right) dx &= \int_0^1 \left(y \frac{\sin x}{x} \right) \Big|_{y=0}^{y=x} dx = \int_0^1 \sin x \, dx \\ &= -\cos(1) + 1 \approx 0.46. \end{aligned}$$

If we reverse the order of integration and attempt to calculate

$$\int_0^1 \int_y^1 \frac{\sin x}{x} dx \, dy,$$

we run into a problem because $\int ((\sin x)/x) dx$ cannot be expressed in terms of elementary functions (there is no simple antiderivative).

There is no general rule for predicting which order of integration will be the good one in circumstances like these. If the order you first choose doesn't work, try the other. Sometimes neither order will work, and then we need to use numerical approximations. ■

Finding Limits of Integration

We now give a procedure for finding limits of integration that applies for many regions in the plane. Regions that are more complicated, and for which this procedure fails, can often be split up into pieces on which the procedure works.

Using Vertical Cross-sections When faced with evaluating $\iint_R f(x, y) dA$, integrating first with respect to y and then with respect to x , do the following three steps:

1. **Sketch.** Sketch the region of integration and label the bounding curves (Figure 15.14a).
2. **Find the y -limits of integration.** Imagine a vertical line L cutting through R in the direction of increasing y . Mark the y -values where L enters and leaves. These are the y -limits of integration and are usually functions of x (instead of constants) (Figure 15.14b).
3. **Find the x -limits of integration.** Choose x -limits that include all the vertical lines through R . The integral shown here (see Figure 15.14c) is

$$\iint_R f(x, y) dA = \int_{x=0}^{x=1} \int_{y=1-x}^{y=\sqrt{1-x^2}} f(x, y) dy \, dx.$$

Using Horizontal Cross-sections To evaluate the same double integral as an iterated integral with the order of integration reversed, use horizontal lines instead of vertical lines in Steps 2 and 3 (see Figure 15.15). The integral is

$$\iint_R f(x, y) dA = \int_0^1 \int_{1-y}^{\sqrt{1-y^2}} f(x, y) dx \, dy.$$

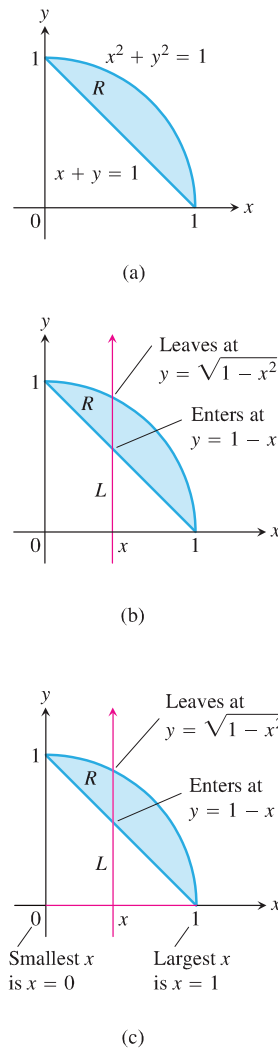


FIGURE 15.14 Finding the limits of integration when integrating first with respect to y and then with respect to x .

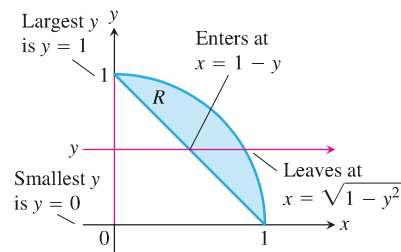


FIGURE 15.15 Finding the limits of integration when integrating first with respect to x and then with respect to y .